Zishuo Zhao

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LLM Agents & Reinforcement Learning

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Education

Sun Yat-sen University *Sep 2021 - Jun 2024*

Master's degree in Intelligent Control

School of Intelligent Systems Engineering, Shenzhen, China

Sun Yat-sen University *Sep 2017 - Jun 2021*

Bachelor's degree in Computational Materials

School of Materials, Guangzhou, China

Work Experience

Alibaba Holding *Aug 3, 2026 - Present*

Large Language Model Algorithm Engineer

- *Code Agent Data Synthesis:* Develop synthetic training data for code agents, focusing on Terminal-Bench 3.0, machine learning, and scientific tasks. Work on task-oriented data and agent interaction trajectories to support coding, experimentation, and problem solving across these domains.
- *Multimodal Agent Training:* Train multimodal models as agents, covering reflection, OPD, tool calling, and revisiting prior observations. Work on reinforcement learning for multimodal agent tasks that combine perception, reasoning, and interaction across multiple steps.
- *Research on Agent Learning:* Investigate Claude's reasoning trajectories through probing and reverse engineering. Explore knowledge distillation, OPD, RSI-based self-improvement, and agentic reinforcement learning, with a focus on reasoning transfer and iterative agent improvement.

Alibaba International *Jul 4, 2024 - Jul 3, 2026*

Large Language Model Algorithm Engineer

- *Customer service agents:* Contributed to AliExpress pre-sales and after-sales agents, including planner-reasoner architectures, tool calling, reward evaluation, and alignment with DPO and GRPO.
- *Multimodal review:* Worked on image-based review with multimodal models, OCR, image classification, and preference alignment.
- *Voice agents:* Worked on Lazada inbound and ICBU outbound voice agents, covering planning, knowledge grounding, interruption handling, and order understanding.

Internship Experience

Alibaba International *Dec 2023 - Feb 2024*

Research Intern

Explored RLHF and DPO for customer service language models, reinforcement learning frameworks, and automated model evaluation.

Tencent AI Lab *May 2023 - Oct 2023*

Research Intern

Worked on LLM role-playing, game commentary, and report generation. Participated in research on PPO-based methods for extending model input context.

Publications

[1] **Zishuo Zhao**, Kai Chen, Ao Li, Yuan Liu.
Turn-level Multiscale Density Ratio Estimation for LLM Agents.
arXiv:2609.16760, 2026.

[2] Jingyou Xie, Jiayi Kuang, Zhenzhou Lin, Jiarui Ouyang, **Zishuo Zhao**, Ying Shen.
FLEX-CLIP: Feature-Level GEneration Network Enhanced CLIP for X-shot Cross-modal Retrieval.
arXiv:2411.17454, 2024.

[3] **Zishuo Zhao**, Ziyang Ma, Zhenzhou Lin, Jingyou Xie, Yinghui Li, Ying Shen.
Source-free Domain Adaptation for Aspect-based Sentiment Analysis.
LREC-COLING 2024, pp. 15076-15086.

[4] Zhenzhou Lin, **Zishuo Zhao**, Jingyou Xie, Ying Shen.
Hierarchical Type Enhanced Negative Sampling for Knowledge Graph Embedding.
ACM SIGIR 2023, pp. 2047-2051.

[5] **Zishuo Zhao**, Yuexiang Xie, Jingyou Xie, Zhenzhou Lin, Yaliang Li, Ying Shen.
Source-Free Unsupervised Domain Adaptation for Question Answering.
IEEE ICASSP 2023, pp. 1-5.

[6] Jingyou Xie, **Zishuo Zhao**, Zhenzhou Lin, Ying Shen.
Multimodal Graph Learning for Cross-Modal Retrieval.
SIAM SDM 2023, pp. 145-153.

Technical Skills

Programming and frameworks: Python, verl.
Methods: DPO, GRPO, RLHF, LLM agent development, and multimodal learning.
Languages: English (CET-6, CET-4).

Research Interests

My research focuses on **large language model post-training and agents**, including OPD, synthetic data generation for code agents, RSI-based self-improvement, and multimodal agents.

- **Agentic reinforcement learning:** Multi-turn preference learning, turn-level credit assignment, and alignment for LLM agents.
- **Natural language processing:** Source-free domain adaptation for question answering and aspect-based sentiment analysis.
- **Multimodal learning:** Cross-modal retrieval, multimodal graph learning, and vision-language applications.